

B. Tech Degree V Semester Examination, November 2009

ME 505 POWER PLANT ENGINEERING (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer all questions)

(8 x 5 =40)

- I.
 - a. Indicate the significance of incremental rate of a power plant.
 - b. What is the protection provided to a power plant against the consumers having high maximum demand with poor load factor.
 - c. What are the factors considered in selecting a type of dam?
 - d. Write notes on various power house equipments.
 - e. Explain the principle of FBC system with a neat sketch.
 - f. Define the term 'vacuum efficiency' applied to a condenser.
 - g. Write short notes on 'solar thermal collection'.
 - h. Sketch the arrangement made in wind mill for speed control.

PART B

(4 x 15 =60)

- II.
 - a. A small power plant capital cost is Rs.40,000 and it's salvage value is Rs.4000/- at the end of a useful life of 20 years. Find it's value half way through it's life based on (a) straight line depreciation method (b) reducing balance depreciation method (c) sinking fund depreciation at 6% compounded annually. (10)
 - b. List the various costs involved in determining the economics of power plant. Plot variation of costs of power plant vs it's capacity. (5)

OR

- III.
 - a. The daily load curve of a power plant is given by the following equation:

$$L = 350 + 10t - t^2$$
 Where t is time in hours from 0 to 24 hrs and L is MW. Draw load curve and load duration curve and calculate
 - (i) Value of maximum load and when it occurs
 - (ii) load factor of the plant. (9)

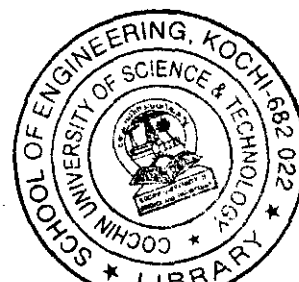
- b. Take the following data for a thermal power plant of 500 MW capacity.

Capacity factor	= 0.45
Annual load factor	= 0.6
Cost of fuel used/year	= Rs.1000 x 10 ⁶
Capital cost of the plant	= Rs.10000 x 10 ⁶
Annual interest and depreciation	= 15%

- Calculate (i) Minimum reserve capacity of the station (6)
- (ii) The cost per KWh generated

- IV.
 - a. Differentiate between a conduit and penstock. How length of penstock is decided in a hydro power plant? What are the advantages and disadvantages of exposed penstock over buried penstock? (7)
 - b. Why the starting of diesel power plant is more difficult? What different methods are used for starting diesel engine? Which method is common and why? (8)

OR



(Turn over)

- V. a. Explain the functions of surge tank. How will you differentiate differential type with that of simple or restricted orifice type? (7)
- b. What do you understand by a closed cycle gas turbine plant? List out its advantages over open cycle plant. What difficulties are encountered in the development of closed cycle plant? (8)
- VI. a. Explain the various ash handling systems. Discuss the relative merits and demerits. (8)
- b. What is chain reaction? How it is useful in nuclear power generation? What are the effects of uncontrolled chain reaction? Explain with examples. (7)
- OR**
- VII. a. Explain the principle used in forced and induced draught. Why balanced draught is preferred over these two. (7)
- b. What factors are mainly considered in the design of boilers used in power plants? Discuss the significance of each with details. (8)
- VIII. a. Describe an MHD closed cycle system, with its advantages and disadvantages. (8)
- b. Explain the methods of ocean thermal electric power generation. (7)
- OR**
- IX. a. Explain the design and principle of operation of a fuel cell with special reference to H_2 , O_2 cell. (8)
- b. Describe the principle of working and constructional details of a basic thermionic generator. (7)
